Russia: National Innovation Strategy Brief

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Country assessment: Russia's national innovation system

Russia is a scientifically capable country whose innovation system underperforms its human capital. In WIPO's Global Innovation Index (GII) 2025, Russia ranks 60th, far behind the United States (3rd), China (10th), and France (13th), and it has fallen from 45th in 2021 [2]. Its profile is lopsided: human capital and research is its best pillar (28th), while institutions rank 131st, by far the weakest of the four game countries. Russia also ranks higher on innovation outputs (55th) than on inputs (73rd), meaning it gets comparatively more out of what it puts in [2].

Indicator	Russia	United States	China	France
GII 2025 overall rank [2]	60	3	10	13
Human capital and research, rank [2]	28	13	20	15
Knowledge and technology outputs, rank [2]	62	3	1	15
Institutions, rank [2]	131	16	44	33
R&D spending, % of GDP [3]	1.10 (2020)	3.59 (2022)	2.56 (2022)	2.22 (2021)
Education spending, % of GDP [1]	4.2 (2023)	5.4 (2021)	4.0 (2023)	5.3 (2022)
Real GDP, PPP, $ trillion, 2024 [1]	6.09	25.68	33.60	3.73

Education. Russia's educational base is broad: literacy is 99.9% and school life expectancy is 15 years [1]. Education spending (4.2% of GDP) is close to China's but below the United States and France. Human capital is the one dimension where Russia is within range of its rivals. Spending shares measure inputs, not quality, and the years differ across countries.

Technology. Russia spends about 1.1% of GDP on R&D, roughly a third of the U.S. rate and less than half of China's or France's [3]. Its knowledge and technology outputs rank 62nd, against China's 1st [2]. Capability is concentrated in legacy sectors such as space, where Russia runs one of the world's largest programs [1]. EU export restrictions on dual-use and advanced technologies further limit access to foreign components [4]. The game reflects this legacy: Russia starts with the most tacit knowledge (TK 7, versus France's 4 and 0 for the United States and China) but the joint-smallest budget (wealth W = 4, or 8 dice per round) [8].

Intelligence and immigration. Russia has a sophisticated state intelligence apparatus; the UK government attributes extensive cyber operations to military intelligence (GRU) units such as Unit 26165 [5]. Its main talent problem is skilled emigration: about 100,000 IT specialists, roughly 10% of the sector, left in 2022 by the government's own count [6]. Aggregate net migration is still slightly positive (+0.76 per 1,000) [1], so population totals hide this outflow. The game builds in the same pattern. Russia has the smallest talent pool (K = 5) and a talent flow of 0.60, so it loses 40% of its talent every round, but its closed borders mean rivals can never steal its points. The United States and France are open and can be robbed [8].

Implication. Russia cannot win a spending race at the start, and it cannot hold on to talent. Its assets are know-how and security. Because the score multiplies weighted technology points by a TK multiplier [8], Russia should grow its budget first, then convert it into high-level research that its TK makes more likely to succeed.

Education recommendation (Tiger)

Russia's real-world strength is breadth of education: near-universal literacy, 15 years of expected schooling, and a human-capital-and-research ranking (28th) far above its overall innovation ranking (60th) [1, 2]. The weakness is retention and use. Skilled workers leave, and weak institutions limit how well trained people are used. The game captures this. Education raises talent (K), but Russia's talent flow of 0.60 multiplies K every round, including talent we just bought. One tertiary-education purchase (+1 K while our borders are closed) shrinks to 0.6 K after one round and 0.1 K after four [8].

I recommend no education spending in Round 1. K matters only at thresholds: each 10 K adds one extra basic-research success, and K below 7 (for M research) or below 12 (for H research) makes our TK count 10 lower [8]. Russia starts at K = 5. Even if we bought both secondary (+0.87 K) and tertiary (+1 K) education every round, the 0.60 drain would hold K near 3, so education cannot reach any threshold. Each die is worth more saved, because a saved die adds 2/3 to wealth and so 4/3 extra dice in every later round [8]. The United States' talent grows 79% a round; Russia's education is a leaky bucket.

This departs from real-world policy, and it is deliberate. A real government should protect a strong education base, but in the game our talent drains faster than we can refill it. The rulebook asks what a country can do when its talent drains away [8]. Our answer is to substitute know-how for talent: every applied-research die and every basic-research success adds TK, which raises our odds at M and H and multiplies our score. I will watch our talent-flow number each round. It depends on wealth and university strength relative to rivals, so if saving pushes it close to 1.0, buying both upgrades each round (2 dice) could lift K above 7 within about three rounds and remove the M penalty. Until then, education dice go to savings and research.

Technology strategy recommendation (Steve)

Russia's real-world technology base is narrow but deep. R&D intensity is about 1.1% of GDP, far below every rival [3], and EU export controls cut off many advanced inputs [4]. Moscow's response has been to concentrate state resources in legacy strengths such as space, nuclear, and defense. GII 2025 still shows Russia producing more output than its inputs predict [2]. The game rewards the same concentration: points count 1, 2, and 3 times for L, M, and H research, and Russia's TK lead makes the big M and H rewards more likely [8].

I recommend buying only the two cheapest applied-research dice in Rounds 1 and 2, at level L. Applied-research costs rise within a round (1, 1, 2, 2, 3, 3 dice) and reset each round, so the first two dice are the best value in the game. Each L die returns about 1.46 points and adds 1 TK even if it fails [8]. We should not buy basic research yet. M research needs 8 effective basic research and H needs 20, and every rival's basic research counts 0.75 toward ours, so rivals' early spending moves us toward the unlocks for free. Our own basic research would also give each rival 0.75 of its value [8].

From Round 3, when saving has quadrupled our budget, we should buy basic research up to the unlock thresholds and then fill the applied-research slots each round (up to 3 L, 2 M, and 1 H die). With TK around 25, an M die returns about 2.8 points and an H die about 1.9; weighted, each is worth about 5.6, nearly four times an L die. Because our K stays below 12, our TK counts 10 lower at H, so H keeps improving as our TK climbs. Our autocratic regime also adds 10% to M and H successes [8]. Gilli and Gilli argue that complex technologies depend on tacit, organizational knowledge that cannot simply be copied [7]; the game rewards that kind of knowledge, and Russia starts with the most.

Intelligence and immigration recommendation (Gloria)

In the real world, Russia combines strong intelligence capabilities with a talent-retention problem. The UK government attributes sophisticated cyber operations to GRU units [5], while about 100,000 IT specialists left in 2022 [6]. Russia's weak institutions (131st) make it hard to attract or win back skilled workers [2]. Russia's real posture is high research security and restricted openness, and the game starts it in the same place: closed borders [8].

I recommend keeping immigration closed for the whole game. At Russia's current university strength, opening would not change our talent flow; the rulebook shows a change of zero for every country if Russia opens. It would only raise the tertiary-education bonus from +1 to +2 K, which drains away anyway, and it would let rivals steal our points with a one-in-three chance per spy [8]. Closed countries can never be robbed. Openness pays for a rich country with strong universities like the United States; for Russia it is all risk.

For espionage, I recommend no spies in Round 1 and two spies per round afterward, at 1 die each. A steal copies the target's points at one level as of the start of the round, so there is nothing to take in Round 1. Only open countries can be robbed. Against the United States or France, a spy steals one time in three and observes one time in six, and a spy on L research is never caught. Spies on M or H risk exposure (one in six and one in three), but a steal there is worth far more: an H steal could bring 36 or more points, counted three times [8]. From Round 2 we should spy at L on the United States and France, then move to M or H once a rival's public basic research shows it can reach those levels. China is closed, so spying on it can never steal and only risks exposure. Because our borders are closed, a caught spy cannot be answered with theft.

Executive recommendation (Steve)

Decision rule. Each czar proposes the allocation for their portfolio. All three members vote, and two votes adopt a decision. As executive, I break ties and decide if the round timer (20 minutes in Round 1, 10 minutes later) is about to expire. Note-taking and statistics duties rotate among all three members, as the instructor asked.

Overall strategy and Round 1 allocation. Our strategy is "save, then strike." Two-thirds of every unspent die becomes wealth at the start of the next round, and the budget is twice wealth, so saving three-quarters of the budget roughly doubles Russia's budget each round [8]. In Round 1 we will spend 2 of our 8 dice on two L applied-research dice, buy no basic research, education, or espionage, keep immigration closed, and save 6 dice. That raises W from 4 to 8 and our Round 2 budget to 16 dice. Round 2 repeats the pattern and adds two L-level spies on the United States and France. From Round 3 we spend: basic research to unlock M and H, full applied research, and spies at higher levels. In our simulation of the rulebook, saving three-quarters of the budget in Rounds 1 and 2 raised Russia's average expected score by roughly 15-40% for games of 3-7 rounds, compared with spending everything each round. The cost is a lower score, by up to about a quarter, if the game ends after only three rounds.

Risks, adjustments, and consensus. Savings in the final round are lost, and the last round is not announced [8], so after Round 2 we will save only dice that research and spies cannot use well. Our rising wealth is public, so rivals may copy the strategy or see us as a threat, but because we are closed they cannot rob us. We will start spending earlier if rivals' basic research unlocks M quickly or if the instructor signals a short game. **[Complete after the team meeting: state whether all three members agreed on each recommendation, any dissent, and how the voting rule resolved it.]**

References

[1] Central Intelligence Agency. *The World Factbook*, entries for Russia, United States, China, and France (archived data, reproduced at theworldfactbook.org after the CIA retired the publication in February 2026). Accessed November 2026.

[2] World Intellectual Property Organization. *Global Innovation Index 2025* country profiles: Russian Federation, United States, China, France.

[3] National Science Board. *Science and Engineering Indicators 2025* (NSB-2025-7), Table DISC-10: GERD, GDP, and GERD-to-GDP ratio by selected country. Russia's latest reported year is 2020.

[4] European Commission. *Export-related restrictions for dual-use goods and advanced technologies*, FAQ on Council Regulation (EU) No 833/2014, updated September 25, 2026.

[5] UK Government. *Profile: GRU cyber and hybrid threat operations*, updated July 13, 2026.

[6] Interfax. "About 100,000 IT specialists left Russia in 2022 - digital development minister," December 20, 2022.

[7] Gilli, Andrea, and Mauro Gilli. 2019. "Why China Has Not Caught Up Yet: Military-Technological Superiority and the Limits of Imitation, Reverse Engineering, and Cyber Espionage." *International Security* 43(3): 141-189. doi:10.1162/isec_a_00337.

[8] Canfil, Justin K. *Global Technology Competition: Official Rulebook*, executive briefing video, and player guide, course materials, November 2026.